

A dual linear programming bound for sphere packing in dimension 36

Rifat Jumagulov
jum.rifm@gmail.com

Draft of July 11, 2026

Abstract

We construct an explicit dual-feasible point for the Cohn–Elkies linear program in dimension 36, built from the space of weight-18 modular forms for $\Gamma_0(24)$ following the method of Cohn and Triantafyllou. The certificate shows that the two-point linear programming bound on the sphere packing density in dimension 36 exceeds the density of the best packing currently known — the Kschischang–Pasupathy packing, of center density $2^{18}/3^{10}$ — by a factor of at least 32.91. In particular, no Cohn–Elkies auxiliary function can certify the best known packing in dimension 36 as optimal. To our knowledge this is the first such dual bound in any dimension above 32, extending the table of Cohn–Triantafyllou ($d = 12, 16, 20, 28, 32$), Li ($3 \leq d \leq 13$), and de Courcy-Ireland–Dostert–Viazovska ($d = 6$). The certificate is exact: the dual point is a rational vector, coefficient nonnegativity is verified by exact arithmetic up to $n = 800$, and eventual positivity of the two relevant q -expansions is proved via an explicit Deligne-type tail bound whose constant is certified with outward-rounded interval arithmetic. Two methodological points may be of independent interest: a constraint-generation (cutting-plane) formulation of the exact rational LP, which is what makes the tail-safe optimum reachable; and a sharpened, lift-aware form of the Deligne bookkeeping constant, $C = \sum_{f,e} |\lambda_{f,e}| e^{-(k-1)/2}$, without which the finite verification in dimension 36 fails marginally.

1 Introduction

The Cohn–Elkies linear programming bound [1] is the strongest known general upper bound on sphere packing densities in moderate dimensions, and it is famously sharp in dimensions 8 and 24 [2, 3]. In every other dimension its sharpness is open, and the question splits into two inequivalent problems. Write $\delta_{\text{LP}}(d)$ for the optimum of the Cohn–Elkies program (normalized as a center density in the sense of [13]) and $\delta_*(d)$ for the best packing density currently known.

- A *dual bound* is a certified lower bound $\delta_{\text{LP}}(d) \geq L$. If $L > \delta_*(d)$, then no Cohn–Elkies function can prove the optimality of the best known packing: either the LP bound is not sharp, or a denser packing than currently known exists. Cohn and Triantafyllou [4] constructed such points from modular forms in dimensions 12, 16, 20, 28, 32; Li [5] gave a different (discrete-reduction) construction covering $3 \leq d \leq 13$; de Courcy-Ireland, Dostert and Viazovska [6] extended the modular-form method to dimension 6.
- *Strict non-sharpness* is the stronger statement $\delta_{\text{LP}}(d) > \delta_{\text{max}}(d)$, where δ_{max} is the true optimal density. This requires, in addition to a dual lower bound $L \leq \delta_{\text{LP}}(d)$, an *independent* upper bound $U \geq \delta_{\text{max}}(d)$ with $U < L$. At present this is known only for

$d \in \{3, 4, 5, 6, 12, 16\}$: for $d = 12, 16$ by combining the Cohn–Triantafillou dual points with the three-point bounds of Cohn–de Laat–Salmon [7], and for low dimensions by Li’s dual bounds against classical upper bounds [5, 6].

This paper proves a statement of the first kind in dimension 36, the first above 32.

Theorem 1.1. $\delta_{\text{LP}}(36) \geq 146.1036\dots = 32.9104\dots \times \frac{2^{18}}{3^{10}}$. *In particular the Cohn–Elkies linear programming bound in dimension 36 cannot certify the Kschischang–Pasupathy packing [8], of center density $2^{18}/3^{10} = 4.4394\dots$, as an optimal sphere packing.*

The witness is an explicit dual-feasible pair of modular forms $(g, \tilde{g})$ of weight 18 for $\Gamma_0(24)$, with $\tilde{g}$ the Fricke transform of g ; the required nonnegativity of the two q -expansions is verified exactly for $1 \leq n \leq 800$ and proved for all $n > n_0$ with an explicit crossover $n_0 \leq 63$, closing the gap.

Dimension 36 is a natural target: the weight $k = d/2 = 18$ is even, so no character twists are needed, and 36 is the first dimension beyond the Cohn–Triantafillou table where $\dim S_{d/2}$ obstructions do not apply. It is also qualitatively different from the dimensions ≤ 32 : the ratio $\delta_{\text{LP}}/\delta_*$ is much larger (our certificate alone gives 32.9; the two-point optimum is $\approx 52 \times \delta_*$), reflecting how weak the record packing is relative to the LP bound in this range. Ironically this makes the *dual* statement stronger and the *strict* question harder; see §5.

Why the construction is not routine. Running the Cohn–Triantafillou method at $d = 36$ meets two obstacles that do not occur (or are mild) at $d \leq 32$.

First, the naïve LP relaxation (maximize the value subject to nonnegativity of finitely many coefficients) produces optima whose q -expansions turn negative just past the enforced range, on the residue classes $n \equiv 3, 9, 15, 21 \pmod{24}$; while the fully constrained problem (nonnegativity through the eventual-positivity crossover, hundreds of rows with coefficients of size $\sim n^{17}$) is numerically singular for floating-point solvers and prohibitively large for exact dense simplex. We resolve this by *constraint generation*: solve the small relaxation exactly in rational arithmetic, evaluate all violated rows exactly, add them, and re-solve. In dimension 36 this converges in a single round (all violations lie in the four classes above and are eliminated together), producing an exact rational vertex whose q -expansions are nonnegative far beyond the enforced range (§3).

Second, the eventual-positivity verification of [4, §5] requires an explicit constant C with $|c_n| \leq C \sigma_0(n) n^{(k-1)/2}$ for the cuspidal parts of *both* g and $\tilde{g}$. The natural constant — the ℓ^1 -norm $\sum |\lambda_{f,e}|$ of the coordinates in the newform-lift basis — turns out to be *lossy* in a way that matters here: our certificate’s cuspidal part is dominated (99.7% of the norm) by a single deep oldform lift, the $e = 24$ shift of the level-one newform, and the naïve constant fails the required threshold marginally. The lift-aware constant $C_w = \sum_{f,e} |\lambda_{f,e}| e^{-(k-1)/2}$ (Lemma 4.1), equally rigorous and elementary, is smaller here by ten orders of magnitude and closes the verification with a margin of 7×10^8 (§4).

A further point, absent from dimensions where the certificate has a sparse structure, is that *both* cusps genuinely matter: not only the Fricke transform $\tilde{g}$ but also g itself has a razor-thin Eisenstein component ($e_1(g) \approx -3.9 \times 10^{-11}$), so the $a_n \geq 0$ side needs the same Deligne-type argument as the $b_n \geq 0$ side, with its own constants (§4).

Rigor. The dual vertex is an exact rational vector (the entries have ≈ 128 -digit numerators). All coefficient scans up to $n = 800$ (and, in a second implementation, to $n = 1100$) are exact.

The newform decompositions of both cuspidal parts are validated by an exact identity: the reconstruction of the q -expansion from the computed coordinates agrees with the exact coefficients *as rational numbers* (deviation identically 0) up to $n = 800$. The tail constants C_w are certified as upper bounds using outward-rounded interval arithmetic, with the algebraic Hecke eigenvalues enclosed by exact Sturm isolating intervals. Independent implementations (exact characteristic-polynomial factorization vs. high-precision simultaneous diagonalization) produce identical constants. All code, receipts, and the exact certificate data are provided as ancillary files (see the manifest in `ancillary/MANIFEST.md`).

2 The Cohn–Elkies program and its modular-form dual

The engine is the following observation of Cohn and Triantafyllou [4, Prop. 2.1]: if μ is a tempered distribution on $\mathbb{R}^d$ with $\mu = \delta_0 + \nu$, $\nu \geq 0$, $\text{supp}(\nu) \subseteq \{x \in \mathbb{R}^d : |x| \geq r\}$ for some $r > 0$, and $\hat{\mu} \geq c\delta_0$ for some $c > 0$, then the Cohn–Elkies linear programming bound in $\mathbb{R}^d$ is at least $c(r/2)^d$. (Any Cohn–Elkies auxiliary function f pairs with μ on both sides of the inequality chain, forcing $f(0)/\hat{f}(0) \geq cr^d$ up to normalization.) Sections 2–3 of [4] construct such distributions from modular forms, as follows.

Fix the dimension $d = 36$, weight $k = d/2 = 18$, and level $N = 24$. For a holomorphic modular form $g \in M_k(\Gamma_0(N))$ write $g = \sum_{n \geq 0} a_n q^n$ ($q = e^{2\pi iz}$) and let

$$\tilde{g} = i^k (g|_k w_N), \quad w_N = \begin{pmatrix} 0 & -1 \\ N & 0 \end{pmatrix}, \quad (f|_k M)(z) = (\det M)^{k/2} (cz + d)^{-k} f(Mz),$$

be its (normalized) Fricke transform, i.e. $\tilde{g}(z) = i^k N^{-k/2} z^{-k} g(-1/(Nz))$; for our even weight $k = 18$ the factor is $i^{18} = -1$. Then $\tilde{g} \in M_k(\Gamma_0(N))$ as well, and $\tilde{\tilde{g}} = g$. Write $\tilde{g} = \sum_{n \geq 0} b_n q^n$. Given g satisfying the coefficient conditions below, the construction of [4, §§2–3] produces a distribution μ as above with $r = \sqrt{T}$ and $c = (2/\sqrt{N})^{d/2} b_0$. Concretely: from any g satisfying

$$a_0 = 1, \quad a_n = 0 \ (1 \leq n \leq T-1), \quad a_n \geq 0 \ (n \geq T), \quad b_n \geq 0 \ (n \geq 0), \quad (1)$$

a dual-feasible point for the Cohn–Elkies program with value (as a center density)

$$B(g) = b_0 \cdot \left(\frac{2}{\sqrt{N}}\right)^{d/2} \cdot \left(\frac{\sqrt{T}}{2}\right)^d, \quad (2)$$

so that $\delta_{\text{LP}}(d) \geq B(g)$. Here T is a free integer parameter (the number of forced initial zeros). We take $T = 10$. As a consistency check of the normalization, our implementation of (1)–(2) reproduces the published $d = 12$ values of [4] at both levels: $0.059782 = 1.6141 \times \text{Coxeter–Todd}$ at $N = 24$ ([4, Table 6.1]) and $0.0624462 = 1.68605 \times$ at $N = 96$ ([4, Thm. 1.1]), to the displayed precision.

3 The certificate

3.1 The space and the reduced LP

$\dim M_{18}(\Gamma_0(24)) = 72$, with an 8-dimensional Eisenstein subspace spanned by $E_{18}(\delta z)$, $\delta \mid 24$, where

$$E_{18}(z) = 1 + \kappa_{18} \sum_{n \geq 1} \sigma_{17}(n) q^n, \quad \kappa_{18} = -\frac{2k}{B_{18}} \Big|_{k=18} = -\frac{28728}{43867},$$

and a 64-dimensional cuspidal subspace. A spanning set of 72 linearly independent forms is obtained from the eight Eisenstein series together with holomorphic eta-quotients of weight 18 and level dividing 24 (Ligozat’s criteria); all q -expansions are computed in exact integer arithmetic. Imposing the 9 linear conditions $a_1 = \dots = a_9 = 0$ and the normalization $a_0 = 1$ leaves an affine space of dimension $r = 20$, on which we solve

$$\max b_0 \quad \text{s.t.} \quad a_n \geq 0 \ (10 \leq n \leq 28), \quad b_n \geq 0 \ (n \in W), \quad (3)$$

with a working set W of Fricke-side rows, in exact rational arithmetic.

3.2 Constraint generation

With $W = \{1, \dots, 28\}$ the exact optimum of (3) is $36.3470 \times 2^{18}/3^{10}$, but its Fricke expansion has 45 negative coefficients in $1 \leq n \leq 300$, all in the classes $n \equiv 3, 9, 15, 21 \pmod{24}$, the first at $n = 33$: the unconstrained optimum overshoots into infeasibility, and (as we show below) its Eisenstein data forces $b_n < 0$ for infinitely many n , so no finite verification can rescue it. Adding all 45 violated rows to W ($|W| = 73$) and re-solving exactly gives a vertex with

$$b_0 = 101.181761 \dots \text{ (exact rational)}, \quad B = 146.10367 \dots = 32.9104 \dots \times \frac{2^{18}}{3^{10}},$$

and *no* violations: re-scanning exactly, $a_n \geq 0$ and $b_n \geq 0$ for all $1 \leq n \leq 800$ (zero negative coefficients on either side, including the un-enforced range $300 < n \leq 800$). The cutting-plane process converges in one round. Exact rational arithmetic in the solve is essential: at this conditioning, floating-point and even 300-digit fixed-precision simplex return phantom vertices (violating $a_0 = 1$) once rows with coefficients of size 10^{30} enter the basis.

Remark 3.1. The certificate data — the 20 exact rational coordinates of the vertex (in the reduced basis of the construction), the exact Eisenstein coefficients, and all constants below — are in the ancillary file `ancillary/certificate_exact_data.txt`; the construction and verification receipts are `ancillary/receipt_d36.cutgen.txt` and `ancillary/receipt_d36.cs.txt`.

4 Eventual positivity

It remains to prove $a_n \geq 0$ and $b_n \geq 0$ for all $n > 800$. Decompose both forms into Eisenstein and cuspidal parts; we treat $\tilde{g}$ (the binding side) and note the same argument for g with its own constants.

4.1 Eisenstein main term, per residue class

Write the Eisenstein component of $\tilde{g}$ as $\sum_{\delta|24} e_\delta E_{18}(\delta z)$. The exact rational e_δ (computed by an exact projector, validated as explained below) have the numerical values

$$\begin{aligned} e_1 &= -4.197922 \cdot 10^{-13}, & e_2 &= -2.023981 \cdot 10^{-14}, \\ e_3 &= +5.361611 \cdot 10^{-5}, & e_4 &= +5.776815 \cdot 10^{-3}, \\ e_6 &= -18.24676, & e_8 &= -27.55171, \\ e_{12} &= +43.87748, & e_{24} &= +103.0969. \end{aligned}$$

For $n \geq 1$ the Eisenstein coefficient is $\kappa_{18} \sum_{\delta|\gamma} e_\delta \sigma_{17}(n/\delta)$, where $\gamma = \gcd(n, 24)$. Using $m^{17} \leq \sigma_{17}(m) < \zeta(17) m^{17}$ together with the rational bound $\zeta(17) < 1 + 2^{-17} + 2^{-16}/16$ (compare the

sum with $\int_2^\infty x^{-17} dx$), and bounding each term from below according to its sign (note $\kappa_{18} < 0$, so $\kappa_{18}e_\delta > 0$ exactly when $e_\delta < 0$), we get for every $n \geq 1$ in the class γ :

$$\text{Eisenstein}(n) \geq c_E(\gamma) n^{17}, \quad c_E(\gamma) = \sum_{\substack{\delta|\gamma \\ e_\delta < 0}} \frac{\kappa_{18}e_\delta}{\delta^{17}} + \left(1 + 2^{-17} + \frac{2^{-16}}{16}\right) \sum_{\substack{\delta|\gamma \\ e_\delta > 0}} \frac{\kappa_{18}e_\delta}{\delta^{17}}, \quad (4)$$

an exact rational number for each γ . Evaluating exactly: all eight constants $c_E(\gamma)$, $\gamma \mid 24$, are *positive*, with the minimum on the class $\gamma = 3$:

$$c_E := \min_{\gamma|24} c_E(\gamma) = c_E(3) = 3.0197203 \cdot 10^{-15} \quad (\tilde{g}\text{-side}), \quad (5)$$

and analogously $c_E^{(a)} = c_E^{(a)}(3) = 5.0672218 \cdot 10^{-12}$ for g (again the minimum over all eight classes, all positive). The displayed decimals are certified truncations of the exact rationals, which are provided in the ancillary files. The thinness of these constants — forced by the active constraints at the LP optimum — is what makes the tail argument in dimension 36 delicate: the certificate value lives 12–14 orders of magnitude above the Eisenstein margin that protects its tail.

4.2 The cuspidal remainder and a lift-aware Deligne constant

Let $S = \tilde{g} - (\text{Eisenstein part}) = \sum c_n q^n \in S_{18}(\Gamma_0(24))$, a space of dimension 64. The newform/oldform basis consists of the lifts $f(ez)$, where f runs over the normalized newforms of level $M \mid 24$ and $e \mid 24/M$; the newspace dimensions are

$$(\dim S_{18}^{\text{new}}(\Gamma_0(M)))_{M|24} = (1, 1, 3, 2, 3, 4, 2, 9) \quad \text{for } M = (1, 2, 3, 4, 6, 8, 12, 24),$$

with lift multiplicities $(8, 6, 4, 4, 3, 2, 2, 1)$, summing to 64. We compute the exact coordinates $\lambda_{f,e}$ of S in this basis (Galois orbits handled by exact characteristic-polynomial factorization of the Hecke action; bad-prime lifts separated by U_2, U_3 and leading q -exponents) and *validate the decomposition exactly*: the reconstruction $\sum \lambda_{f,e} f(ez)$ agrees with S coefficient-by-coefficient, as exact rationals, for all $n \leq 800$ (deviation identically zero; the Sturm bound for $M_{18}(\Gamma_0(24))$ is 72, so agreement to 800 is a highly overdetermined identity). As a further check, the level-one component is the unique normalized cusp form ΔE_6 of weight 18, and the computed eigenvalues match $a_2 = -528$, $a_3 = -4284$, $a_5 = -1025850$, $a_7 = 3225992$ exactly.

Lemma 4.1 (lift-aware Deligne constant). *Let $S = \sum_{f,e} \lambda_{f,e} f(ez) \in S_k(\Gamma_0(N))$ with f normalized newforms. Then for all $n \geq 1$,*

$$|c_n| \leq C_w \sigma_0(n) n^{(k-1)/2}, \quad C_w = \sum_{f,e} \frac{|\lambda_{f,e}|}{e^{(k-1)/2}}.$$

Proof. By Deligne's bound $|a_f(m)| \leq \sigma_0(m) m^{(k-1)/2}$ for each normalized newform. The coefficient of q^n in $f(ez)$ is $a_f(n/e)$ if $e \mid n$ and 0 otherwise, and for $e \mid n$, $\sigma_0(n/e) \leq \sigma_0(n)$ and $(n/e)^{(k-1)/2} = n^{(k-1)/2} e^{-(k-1)/2}$. Summing over the pairs (f, e) with $e \mid n$ gives the claim. $\square$

The point of the lemma is quantitative. The naïve constant $\sum_{f,e} |\lambda_{f,e}|$ equals 1.894×10^{10} for our S : it is dominated (99.7%) by the single coordinate $\lambda_{\Delta E_6, 24} \approx -1.89 \times 10^{10}$ on the deepest

oldform lift, and with it the crossover lands just *beyond* the verified range ($n_0 \approx 827 > 800$) — the verification fails marginally. The lift-aware constant suppresses that term by $24^{17/2} \approx 5.4 \times 10^{11}$:

$$C_w \leq 0.358807778339 \quad (\tilde{g}), \quad C_w^{(a)} \leq 0.306329333530 \quad (g), \quad (6)$$

certified as upper bounds by outward-rounded interval arithmetic: the coordinates on rational (one-dimensional Galois orbit) blocks, including the dominant level-one tower, are computed as exact rational numbers (31 of the 64 lifts); the remaining 33 algebraic coordinates are enclosed via exact Sturm isolating intervals for the Hecke eigenvalues and an interval linear solve (enclosure widths $\sim 10^{-50}$; the bound is stable to 15 digits across four disjoint determining windows, and degrading the working precision widens the enclosure without breaking the bound, confirming outward rounding).

4.3 Closing the certificate

Combining (5), Lemma 4.1, (6) and $\sigma_0(n) \leq 2\sqrt{n}$: for n on any residue class,

$$b_n \geq c_E n^{17} - 2 C_w n^9 > 0 \quad \text{whenever } n^8 > \frac{2C_w}{c_E},$$

i.e. for all $n \geq n_0 = \lceil (2C_w/c_E)^{1/8} \rceil = 63$; and on the g -side for all $n \geq n_0^{(a)} = 25$. Since $a_n \geq 0$ and $b_n \geq 0$ have been verified exactly for $1 \leq n \leq 800$, both hold for all n , and $(g, \tilde{g})$ satisfies (1). Theorem 1.1 follows from (2). $\square$

The margins are enormous ($2C_w/c_E$ would have to exceed its actual value by a factor 7×10^8 before n_0 reached 800); the certificate is in no sense borderline once the correct constant is used.

5 Discussion

Strict non-sharpness in dimension 36 is far out of reach. Our dual point shows $\delta_{\text{LP}}(36) \geq 32.91 \delta_*$, and the two-point optimum itself is $\approx 52.2 \delta_*$ [1, 4]. Strict non-sharpness would need an independent upper bound below $32.91 \delta_*$, i.e. an improvement of $\approx 37\%$ over the two-point bound. For comparison, the only technology that has ever improved on the two-point bound — the three-point bounds of [7] — gains 4.85% at $d = 12$ and 0.79% at $d = 16$, and no three-point computation exists in any dimension beyond 16 [7, 12]. The paradox of dimension 36 is that the *weakness* of the known record makes the dual statement strong and the strict statement inaccessible: the dual value sits far below the LP optimum precisely because the record is far below both.

Dimensions 40 and 44. The same pipeline reaches $d = 40$ (weight 20) in principle, with two new costs: the weight-20 eta-quotients of level 24 span only a 72-dimensional subspace of the 80-dimensional $M_{20}(\Gamma_0(24))$, so the basis must be augmented (e.g. by $E_4 \cdot$ (weight-16 forms)), and the exact arithmetic is heavier (σ_{19} growth). We have not completed a $d = 40$ certificate. We note that Remark 4.3 of [11] asserts that for $d = 32$ and $d = 40$ “LP non-sharpness is established by other means”, without a supporting citation for the case $d = 40$; we are not aware of any dual bound or non-sharpness proof in dimension 40. The modular-bootstrap analysis of [9] (building on the sphere packing–quantum gravity correspondence of [10]) rules out *sharpness* of the LP bound for a range of dimensions by an implied-kissing-number criterion, but reports its first crossings of that criterion only near $d \approx 180$ and, in its authors’ own words, “does not give a

quantitative improvement”; it therefore neither contains nor implies a dual bound at $d = 36$ or 40.

Methodological remarks. Two ingredients may be reusable. (i) Constraint generation appears to be the right formulation of the exact dual LP whenever eventual positivity must be enforced through a crossover: it keeps every exact solve small (here: one re-solve with 73 rows) while the dense exact LP is intractable and floating-point LP is unsound at these conditionings. (ii) Lemma 4.1 is elementary, but the failure of the unweighted constant in dimension 36 suggests it should be the default bookkeeping in eventual-positivity verifications: deep oldform lifts are exactly the coordinates the unweighted ℓ^1 norm overweights, by the factor $e^{(k-1)/2}$ that the lemma restores.

Reproducibility. The exact verification — the rigorous part of the certificate — uses only exact rational linear algebra, integer q -expansions of eta-quotients, and interval arithmetic (Python: `sympy`, `mpmath`, `fractions`); no computer-algebra system with modular-forms machinery is required. Floating-point routines (`numpy`, `scipy`) are used only in the *construction* step, as a heuristic to locate the candidate vertex and basis; they play no role in the exact rigor pass, and every constant reported here is recomputed exactly. The full pipeline, receipts, and the exact certificate are provided as ancillary files, together with an independent second implementation of the newform decomposition (high-precision simultaneous diagonalization) that reproduces the constants (6) to all reported digits.

Data and code availability. The exact certificate data, the construction and verification receipts, and the complete Python pipeline (including the independent second implementation) are provided as ancillary files accompanying this submission; see `ancillary/MANIFEST.md` for the file map and reproduction order.

Acknowledgements. The construction, the exact-arithmetic certificate, and the preparation of this manuscript were carried out with substantial assistance from AI language-model tools. The correctness of the main result does not rest on trust in those tools: the certificate is exact and machine-checkable, and every numerical claim has been reproduced by re-running the exact-arithmetic verification from scratch (the pipeline and receipts are provided as ancillary files). The author is accountable for this work and for the decision to publish it.

Mathematics Subject Classification (2020)

Primary 52C17 (Packing and covering in n dimensions); Secondary 11F11 (Holomorphic modular forms of integral weight), 11F30 (Fourier coefficients of automorphic forms), 90C05 (Linear programming).

Keywords: sphere packing, linear programming bound, Cohn–Elkies bound, dual bound, modular forms, eta-quotients, eventual positivity, exact arithmetic.

References

- [1] H. Cohn and N. Elkies, *New upper bounds on sphere packings I*, Ann. of Math. **157** (2003), 689–714. arXiv:math/0110009.

- [2] M. Viazovska, *The sphere packing problem in dimension 8*, Ann. of Math. **185** (2017), 991–1015.
- [3] H. Cohn, A. Kumar, S. D. Miller, D. Radchenko and M. Viazovska, *The sphere packing problem in dimension 24*, Ann. of Math. **185** (2017), 1017–1033.
- [4] H. Cohn and N. Triantafyllou, *Dual bounds for sphere packing via modular forms*, Math. Comp. **91** (2022), 491–508. arXiv:1909.04772.
- [5] R. Li, *Dual linear programming bounds for sphere packing via discrete reductions*, Adv. Math. **460** (2024), 110043. arXiv:2206.09876.
- [6] M. de Courcy-Ireland, M. Dostert and M. Viazovska, *Six-dimensional sphere packing and linear programming*, Math. Comp. **93** (2024), 1993–2029. arXiv:2211.09044.
- [7] H. Cohn, D. de Laat and A. Salmon, *Three-point bounds for sphere packing*. arXiv:2206.15373.
- [8] F. R. Kschischang and S. Pasupathy, *Some ternary and quaternary codes and associated sphere packings*, IEEE Trans. Inform. Theory **38** (1992), 227–246.
- [9] N. Afkhami-Jeddi, H. Cohn, T. Hartman, D. de Laat and A. Tajdini, *High-dimensional sphere packing and the modular bootstrap*, J. High Energy Phys. (2020). arXiv:2006.02560.
- [10] T. Hartman, D. Mazáč and L. Rastelli, *Sphere packing and quantum gravity*, J. High Energy Phys. (2019). arXiv:1905.01319.
- [11] J. Zhou, *Cusp form dimensions, lattice uniqueness, and LP sharpness for sphere packing in dimensions 8 and 24*. arXiv:2604.10914.
- [12] F. Vallentin, *Conic optimization for extremal geometry*. arXiv:2510.06960.
- [13] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, 3rd ed., Springer, 1999.